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INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Employee Management System

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About me

- **Background :** **MBA in Finance**
- **Motivation :** **Coming from a process executive background in SLA/quality tracking and business documentation, I wanted to move into a role where I work directly with data to drive decisions.**
- **Experience :** **2 years as senior process executive working on SLA/quality tracking, business documentation and advanced MS Office .**
- **LinkedIn and GitHub profile :**
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Agenda

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Business Problem

Organizations often face challenges in managing employee records, payroll, qualifications, and leave data across multiple systems. This makes it difficult to generate accurate reports and gain insights for effective workforce management and decision-making. Manual data management increases the chances of errors, duplicate records, and delays in processing employee information. A well-designed Employee Management System using SQL helps organize data, improve accuracy, simplify reporting, and support faster business decisions

Problem Statement

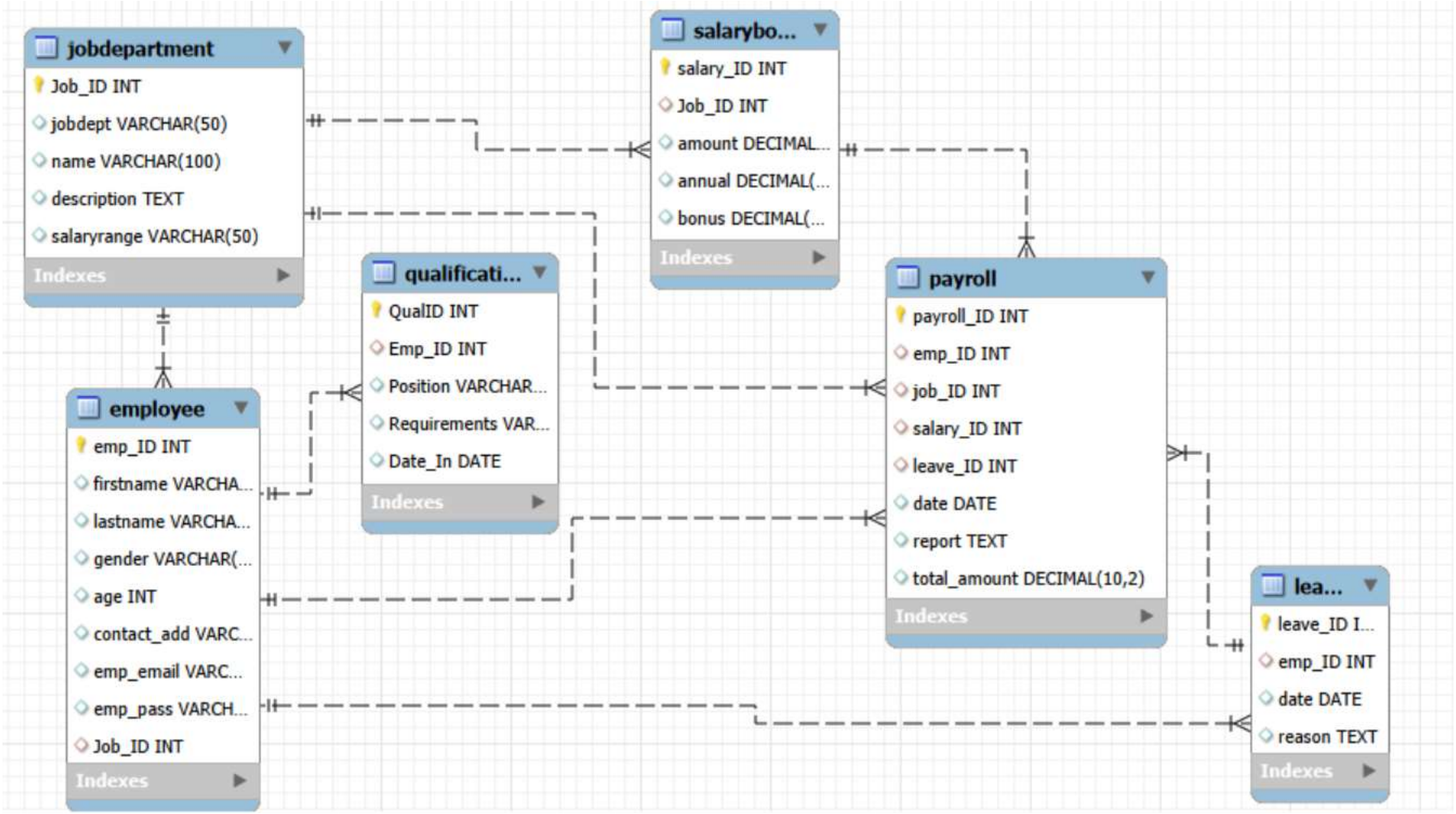
Organizations require a reliable system to manage employee information accurately and efficiently. This project aims to design and implement an Employee Management System using SQL that efficiently stores, organizes, and manages employee-related data within an organization. The system maintains employee details, job roles, salary information, qualifications, leave records, and payroll data in a structured relational database. It aims to improve data accuracy, reduce manual effort, support efficient data retrieval, and generate meaningful reports to help HR and management make informed decisions.

Objective of the Project

- To design and implement a centralized Employee Management System using SQL.
- To store and manage employee information efficiently in a relational database.
- To maintain data accuracy and integrity using primary keys, foreign keys, and constraints.
- To reduce data redundancy and improve consistency across employee records.
- Performed payroll, salary, qualification, and leave analysis to support HR decision-making.
- Applied SQL concepts such as JOINS, Aggregate Functions, GROUP BY, ORDER BY, and Constraints.
- To enhance decision-making by providing accurate and meaningful employee insights.

ER DIAGRAM

The database contains **six interconnected tables**, each performing a specific function.



Database

- JobDepartment Stores job roles, departments, and descriptions.
- SalaryBonus Contains salary, bonus, and annual pay for each job role.
- Employee Holds employee personal information, contact details, and job assignment.
- Qualification Maintains qualification, skills, and requirements for each employee's position.
- Leaves Tracks employee leave details such as date and reason.
- Payroll Combines employee, job, salary, and leave data to calculate final payments.
- Foreign Key Relationships The tables are connected using foreign keys with CASCADE and SET NULL rules to maintain data integrity

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SQL Queries

EMPLOYEE INSIGHTS

Q. How many unique employees are currently in system?

```
select count(distinct emp_id) as total_employees
from employee;
```

	total_employees
▶	60

Key insights :

- The Organization currently has 60 unique employees in the system.
- Each employee record is uniquely identified using emp_ID.
- It also ensures there are no duplicate employee records in the database.

Q. Which departments have the highest number of employees?

```
select jobdept,count(e.emp_ID) as employee_count
from employee as e
join jobdepartment as j
on e.job_ID=j.job_ID
group by j.jobdept
order by employee_count desc;
```

jobdept	employee_count
Finance	9
IT	9
Operations	8
Marketing	8
Engineering	7
Sales	7
HR	7
Legal	5

Key Insights :

- The Finance and IT departments has the highest number of employees with 9 employees each
- The Operations department follows with 8 employees
- Engineering, Sales, and other departments are having 7 employees and less.

Q. What is the average salary per department?

```sql
select j.jobdept,avg(s.amount) as avg_salary
from employee e
join jobdepartment as j
on e.job_id = j.job_id
join salarybonus as s
on j.job_id = s.job_id
group by j.jobdept;
```

jobdept	avg_salary
Operations	68750.000000
Finance	72333.333333
IT	70888.888889
Marketing	65625.000000
Engineering	81142.857143
Sales	75428.571429
HR	62571.428571
Legal	84600.000000

Key Insights :

- The Legal department has the highest average salary at approximately 84,600.
- Engineering follows with an average salary of around 81,143.
- The HR department has the lowest average salary at 62,571.

Q. Who are the top 5 highest-paid employees?

```
select e.firstname,e.lastname,s.amount as salary
from employee e
join salarybonus as s
on e.job_id = s.job_id
order by s.amount desc
limit 5;
```

firstname	lastname	salary
Ingrid	Adams	170000.00
John	Baker	160000.00
Grake	Moor	150000.00
Hank	Wilson	150000.00
Kelly	Cooper	140000.00

Key Insights :

- Ingrid is the highest-paid employee with a salary of 170,000.
- John holds the second-highest salary at 160,000.
- Grake and Hank both earn 150,000.

JOB ROLE AND DEPARTMENT ANALYSIS

Q. How many different job roles exist in each department?

```
select jobdept, count(name)
from Jobdepartment
group by jobdept;
```

jobdept	count(name)
Operations	8
Finance	9
IT	9
Marketing	8
Engineering	7
Sales	7
HR	7
Legal	5

Key Insights :

- The Finance department has the highest number of job roles with 9 roles.
- IT and Operations departments each have 8 job roles.
- The Legal department has the lowest number of job roles with 5 roles.

Q. What is the average salary range per department?

```
select jobdept,avg(amount) as avg_salary
from jobdepartment j
join Salarybonus s
on j.job_id = s.job_id
group by jobdept;
```

jobdept	avg_salary
Operations	68750.000000
Finance	72333.333333
IT	70888.888889
Marketing	65625.000000
Engineering	81142.857143
Sales	75428.571429
HR	62571.428571
Legal	84600.000000

Key Insights :

- The Legal department has the highest average salary range at approximately 84,600.
- Engineering and Sales departments also have high average salary ranges.
- The Finance and IT departments maintain competitive salary structures.
- The HR department has the lowest average salary range at approximately 62,571.

Q. Which job roles offer the highest salary?

```sql
select name,amount as highest_salary
from Jobdepartment j
join Salarybonus s
on j.job_id = s.job_id
order by amount desc
limit 1;
```

name	highest_salary
Finance Director	170000.00

Key Insights :

- The Finance Director role offers the highest salary at 170,000.
- Engineering Director is the second highest-paid role with 160,000.
- Marketing Director and IT Director both offer salaries of 150.000

Q.Which departments have the highest total salary allocation?

```
select jobdept,sum(amount)
from Jobdepartment j
join Salarybonus s
on j.job_id = s.job_id
group by jobdept
order by sum(amount) desc;
```

jobdept	sum(amount)
Finance	651000.00
IT	638000.00
Engineering	568000.00
Operations	550000.00
Sales	528000.00
Marketing	525000.00
HR	438000.00
Legal	423000.00

Key Insights :

- The Finance department has the highest total salary allocation at 651,000.
- The IT department follows with a total allocation of 638,000.
- The Legal department has the lowest total salary allocation at 423,000.

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QUALIFICATION AND SKILLS ANALYSIS

Q. How many employees have at least one qualification listed?

```
select count(distinct emp_id)
from Qualification;
```

count(distinct emp_id)
60

Key Insights :

- All 60 employees in the organization have at least one qualification recorded.
- This indicates that the workforce is fully qualified for their respective job roles.
- The organization maintains proper qualification records for all employees.

Q. Which positions require the most qualifications?

```
select position,count(requirements)
from Qualification
group by position
order by count(requirements) desc;
```

position	count(requirements)
HR Assistant	1
HR Manager	1
IT Support	1
Software Engineer	1
Data Analyst	1

Key Insights :

- All listed job positions currently have one qualification requirement each.
- Positions such as HR Manager, Software Engineer, Data Analyst, and Marketing Manager requires specific qualifications.
- The organization maintains qualification standards across different job roles

LEAVE AND ABSENCE PATTERNS

Q. Which year had the most employees taking leaves?

```
select year(date),count(distinct emp_id)
from leaves
group by year(date)
order by count(*) desc;
```

year(date)	count(distinct emp_id)
2024	60

Key Insights :

- The Year 2024 recorded the highest number of employees leaves with 60 leave entries.
- This indicates increased leave activity during the year.
- The analysis helps HR monitor employee absence trends over time.

Q. What is the average number of leave days taken by its employees per department?

```
select jobdept,avg(leave_count) as avg_leaves
from (
	select emp_id,count(*) as leave_count
	from leaves
	group by emp_id
) l
join employee e
on e.emp_id = l.emp_id
join jobdepartment j
on e.job_id = j.job_id
group by jobdept;
```

jobdept	avg_leaves
Operations	1.0000
Finance	1.0000
IT	1.0000
Marketing	1.0000
Engineering	1.0000
Sales	1.0000
HR	1.0000
Legal	1.0000

Key Insights :

- All departments have an average leave count of 1 leave per employee.
- Leave distribution is consistent across all departments.
- No department shows unusually high or low leave activity

PAYROLL AND COMPENSATION ANALYSIS

Q. What is the average bonus given per department?

```
select jobdept,avg(bonus) as avg_bonus
from jobdepartment j
join Salarybonus s
on j.job_id = s.job_id
group by jobdept;
```

jobdept	avg_bonus
Operations	9687.500000
Finance	10666.666667
IT	10444.444444
Marketing	9125.000000
Engineering	12571.428571
Sales	11214.285714
HR	8171.428571
Legal	13300.000000

Key Insights :

- Legal has the highest average bonus at 13,300.
- HR has the lowest average bonus at 8,171.
- Bonus differ by over 5,000 across teams.

Q. Which department receives the highest total bonuses?

```
select jobdept,sum(bonus) as highest_bonus
from jobdepartment j
join Salarybonus s
on j.job_id = s.job_id
group by jobdept
order by highest_bonus desc;
```

jobdept	highest_bonus
Finance	96000.00
IT	94000.00
Engineering	88000.00
Sales	78500.00
Operations	77500.00
Marketing	73000.00
Legal	66500.00
HR	57200.00

Key Insights :

The Finance department receives the highest total bonus amount at 96.000.

The IT department follows closely with 94,000 in bonuses.

Engineering and Sales departments also receive high bonus allocation.

The HR department has the lowest total bonus amount at 57,200.

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Key Insights

- Identified highest-paid employees and total payroll expenditure.
- Analyzed department-wise salary and bonus distribution.
- Finance department recorded the highest total bonus allocation.
- Evaluated employee qualifications for workforce planning.
- Monitored leave utilization and employee attendance trends.
- Generated department-wise workforce insights.
- Improved reporting efficiency through SQL-based analysis.
- Supported better HR and management decision

Challenges

- Defining correct table relationships and ensuring accurate use of foreign keys.
- Maintaining data consistency with cascading updates and deletes.
- Writing complex joins for reports involving employee roles, leaves, and payroll.
- Ensuring all date fields follow the YYYY-MM-DD format for reliable time-based analysis.
- Preventing duplicate entries using unique constraints, especially on email fields.

Conclusion

The Employee Management System successfully demonstrates how SQL can organize and analyze large employee datasets. The database effectively stores and organizes data related to employees, departments, payroll, bonuses, qualifications, and leave records. SQL queries were used to analyze employee performance, salary distribution, bonus allocation, qualifications, and leave management. The insights generated from the system support better HR management and organizational decision-making. The project helps reduce manual effort, improve reporting efficiency, and maintain centralized employee records. Overall, the Employee Management System demonstrates how database technology can streamline workforce management and support data-driven business operations.

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THANK YOU